Kavya Chowdary Immanni

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FULL STACK ENGINEER | REACT · NODE.JS · TYPESCRIPT · JAVA · SQL

PROFESSIONAL SUMMARY

Full stack engineer with 5 years of experience building production systems end to end with React, TypeScript, Node.js, Java/Spring Boot and SQL. At the University of Iowa, I designed, deployed and support a research submission portal with 60+ REST endpoints and a financial dashboard covering $36M in revenue, from database design to Linux deployment. Before that, I built Java microservices, Kafka pipelines and React screens for Medicare and commercial claims (EDI X12 837) at Optum, with a focus on performance, security and reliability.

TECHNICAL SKILLS

Front End: React, TypeScript, JavaScript (ES6+), Next.js, Redux Toolkit, React Query, Vite, Material-UI, Ant Design, Recharts, Backbone.js, Handlebars, HTML, CSS/SCSS

Back End: Node.js, Express, Java 17, Spring Boot, Spring Data JPA, Hibernate, Spring Security, REST APIs, GraphQL, WebSockets (Socket.IO), Apache Kafka, Python

Databases: Microsoft SQL Server, PostgreSQL, Oracle, MySQL, MongoDB, Redis, SuiteQL, Sequelize, Stored Procedures, Query Tuning

Integrations: Oracle NetSuite (SuiteScript 2.x, RESTlets, User Event Scripts, SuiteFlow), DocuSign API, Celigo, Webhooks, SFTP, EDI X12 837, HL7, FHIR, LLM APIs

Testing & Security: Jest, Mocha, Chai, Playwright, Selenium, JUnit 5, Mockito, Postman, axe-core, OAuth2, JWT, LDAP, bcrypt, Role-Based Access, HIPAA

DevOps & Tools: Linux, systemd, Bash, PowerShell, Docker, Kubernetes (AKS), Azure, AWS, CI/CD (GitHub Actions, Jenkins), Git, Jira, Confluence, Splunk, Grafana

WORK EXPERIENCE

Application Developer – Full Stack — Feb 2026 – Present

Developmental Studies Hybridoma Bank (DSHB), University of Iowa | Iowa City, IA

Tech: React, TypeScript, Node.js, Express, SQL Server, NetSuite, Python, PostgreSQL, Linux, LLM APIs

- Designed, built and deployed the Depositor Portal that research labs worldwide use to submit antibodies to the bank, on React, Node.js/Express and Microsoft SQL Server with a 60+ endpoint REST API, and own it in production.
- Leading the modernization of the antibody intake-to-catalog process: new portal signups create NetSuite CRM records through RESTlets, and NetSuite User Event scripts and SuiteFlow workflows assign catalog IDs when records are created.
- Designed the portal UI as 30 React components with clear layouts and consistent interaction patterns, including a multi-step submission form with drafts and bulk Excel/CSV upload, and added real-time admin notifications over WebSockets (Socket.IO).
- Set up the production environment on an on-prem Linux server with systemd services, Bash and PowerShell release scripts for atomic deploys and rollback, and health checks, and run a self-hosted NocoDB instance in Docker for staff data access.
- Secured the platform with JWT role-based access control, email OTP verification, bcrypt hashing with a zero-downtime migration of old passwords, API rate limiting, and secrets kept in environment configuration.
- Developed a financial analytics dashboard in React on SuiteScript 2.x and SuiteQL covering 190K+ sales orders and $36M in revenue, and cut its load time from about 3 minutes to a few seconds with SuiteQL aggregates, N/cache with cache warming, and server-side pagination.
- Tracked down a caching defect that inflated reported revenue 12 times, following it across the server-side and browser caches, and fixed it so leadership reports showed correct figures again.
- Wrote a Node.js export engine that generates NetSuite import files at roughly 45,000 rows per second, with p95 latency tracked in an automated benchmark suite, so staff no longer re-key product data by hand.
- Own the third-party integrations: a weekly Descartes Visual Compliance screening job over SFTP with encrypted credentials and failure alerts, the DocuSign eSignature API in the storefront, and NetSuite connections to Celigo and Bioz.
- Maintain and extend the live storefront for nearly 5,000 antibodies on a modular SuiteCommerce Advanced extension architecture (Backbone.js, Handlebars, SCSS), and replaced a paid third-party search vendor with native search.
- Set up data pipelines: a Python ETL job that loads 1M+ ERP records into PostgreSQL in under 10 minutes, and a SuiteQL to Google Sheets to Looker Studio feed for leadership reports without manual exports.
- Prototyped an AI product assistant using LLM APIs (Llama via Groq) with tool calling and automatic model fallback that answers researcher questions from a catalog that syncs twice a day.
- Set up the testing practice with 50 Jest unit and API tests (82% coverage), 22 Playwright end-to-end tests with axe-core accessibility checks, and Lighthouse baselines, and gather requirements from lab research staff, office staff and finance.

Software Engineer – Healthcare Claims Platform — Aug 2023 – Jan 2026

Optum, Inc. | Remote

Tech: Java 17, Spring Boot, React, TypeScript, Node.js, Kafka, Redis, EDI 837, Docker, AKS, Splunk

- Developed Java 17 and Spring Boot microservices for Medicare and commercial claims that parse EDI X12 837 transactions through intake, validation, eligibility and adjudication, part of a team effort that cut service latency by 35%.
- Wrote claim validation rules for ICD-10, CPT/HCPCS and NPI codes, and built 15 to 25 REST and GraphQL endpoints for claim status and history with pagination, filtering and OpenAPI documentation.

- Built React UI features with Material-UI and React Hooks for claim search, history and pended-claim work queues, including reusable tables with sorting, filtering and pagination, validated forms, and virtualization for large result sets.
- Built Kafka producers and consumers with idempotency checks, retry topics, dead-letter queues and retrigger handling, so failed claims in high-volume pipelines recovered automatically instead of being reprocessed by hand.
- Tuned JVM settings and used thread dumps, thread pools and asynchronous processing to fix slow claim APIs, contributing to a team-wide 3x increase in API throughput under peak load.
- Secured REST and GraphQL APIs with OAuth2, JWT and Spring Security following OWASP practices, and protected patient data under HIPAA with encryption, role-based access, audit logging and masking in logs and screens.
- Wrote Python scripts for data checks and pipeline support, built NetSuite RESTlet integrations for pharmacy inventory, order and fulfillment records, and mapped HL7 and FHIR data from EHR sources.
- Containerized services with Docker and deployed them to Azure Kubernetes Service (AKS) through Jenkins and GitHub Actions pipelines with health-check-based rollback, with JUnit 5 and Mockito tests on every build.
- Built Splunk and Grafana dashboards with proactive alerting, contributing to a 15% increase in system uptime, and resolved 10 to 15 production incidents a week on the on-call rotation.
- Adopted GitHub Copilot for refactoring and test generation, contributing to a team-wide reduction of about 20% in development cycle time, and built Tableau dashboards on claims data for operational reporting.

Associate Globalization Developer – Internationalization (i18n) & Localization **Aug 2021 – Nov 2022**

TIBCO Software Inc. | Hyderabad, India

Tech: Java, Spring Boot, Spring Data JPA, Oracle, SQL Server, TIBCO EMS/JMS, XLIFF, JavaParser, ESLint, Jenkins, AWS

- Built MOLEE, a localization framework that automates localization of TIBCO software products, reducing average localization turnaround from 3 to 4 weeks down to 4 to 5 days.
- Developed the MOLEE web application from scratch in React, ES6 JavaScript, Alt.js, HTML5 and CSS3 with a Java Spring Boot back end, including statistics dashboards, an online translation review editor, search, pagination and role-based routing.
- Developed Node.js and Express microservices on MySQL with Sequelize, and implemented Gmail and LDAP sign-in with Passport.js so internal teams and translation vendors could use the platform with their existing accounts.
- Integrated Google Neural Machine Translation and Amazon Translate APIs into the pipeline, cutting translation costs from thousands of dollars to a few hundred, and added translation caching to reduce processing time.
- Built a text-to-speech service with Google Cloud TTS and Amazon Polly to localize TIBCO tutorial videos, and implemented i18n for 6 languages with react-intl, cldr.js and moment.js.
- Built TIBCO EMS/JMS message processing with client acknowledgment, error queues and replay so failed localization jobs could be recovered, and sped up large jobs with multithreaded processing and JVM tuning.
- Fixed about 10 to 15 internationalization defects a week with QA, including character encoding, locale formatting and Arabic right-to-left layout issues, and automated builds with Maven, Jenkins and Bash to Docker on AWS.
- Wrote unit tests with Mocha and Chai, API tests with Postman and Chai BDD, and UI automation with Selenium (Java), and built scripts that automated processes and migrated thousands of documents.

PROJECTS

Playverse – 3D Game Hub | React, TypeScript, Vite, Ant Design, Babylon.js, Node.js, Express, Socket.IO, MongoDB, PWA | Live · GitHub

- Built and deployed a full-stack 3D game platform with 4 playable games, accounts, leaderboards and saved progress, installable as a mobile PWA and a React Native Android app, with responsive layouts for desktop and phone.
- Ran real-time multiplayer hide-and-seek on a server-authoritative Socket.IO game server covered by 11 server tests, with enemy AI (vision cones, A* pathfinding) across 5 themed worlds.

EDUCATION

M.S. in Computer Science | GPA: 3.86/4.0 **Dec 2024**

Saint Louis University, St. Louis, Missouri, USA

B.Tech in Computer Science | GPA: 3.68/4.0 **Jul 2021**

Gokaraju Rangaraju Institute of Engineering & Technology, Hyderabad, India